

The primal problem is

$$\min_{x \in \mathbb{R}^2} 4x_1 + 2x_2$$

subject to

$$\begin{aligned} x_1 - x_2 &= 1, \\ x_1 &\geq 0, \\ x_2 &\geq 0. \end{aligned}$$

Rewriting the equality constraint as $x_1 = 1 + x_2$, the problem can be written as

$$\min_{x \in \mathbb{R}^2} 4 + 6x_2$$

subject to

$$\begin{aligned} x_1 &\geq 0, \\ x_2 &\geq 0. \end{aligned}$$

The optimal solution is $x_2^* = 0$ and $x_1^* = 1 + x_2^* = 1$. The optimal value is 4. Consequently, we expect the dual problem to have an optimal solution as well.

To write the Lagrangian, we first write the problem as follows:

$$\min_{x \in \mathbb{R}^2} 4x_1 + 2x_2$$

subject to

$$\begin{aligned} h_1(x) &= 1 - x_1 + x_2 = 0 \quad (\lambda), \\ g_1(x) &= -x_1 \leq 0 \quad (\mu_1), \\ g_2(x) &= -x_2 \leq 0 \quad (\mu_2), \end{aligned}$$

where λ , μ_1 and μ_2 are the penalty parameters associated with each constraint.

The Lagrangian function is

$$\begin{aligned} L(x_1, x_2, \lambda, \mu_1, \mu_2) &= 4x_1 + 2x_2 + \lambda(1 - x_1 + x_2) - \mu_1 x_1 - \mu_2 x_2 \\ &= (4 - \lambda - \mu_1)x_1 + (2 + \lambda - \mu_2)x_2 + \lambda. \end{aligned}$$

The dual function is

$$q(\lambda, \mu_1, \mu_2) = \min_{x_1, x_2} L(x_1, x_2, \lambda, \mu_1, \mu_2).$$

In order for the dual function to be bounded, the coefficients of x_1 and x_2 have to be zero, since we removed all the constraints on x_1 and x_2 . Therefore, we have to impose the following constraints on the penalty parameters:

$$\mu_1 = 4 - \lambda, \tag{1}$$

$$\mu_2 = 2 + \lambda. \tag{2}$$

In that case, the dual function becomes

$$q(\lambda, \mu_1, \mu_2) = \min_{x_1, x_2} \lambda = \lambda.$$

Moreover, the parameters associated with the inequality constraints must be non negative. That is, $\mu_1 \geq 0$ and $\mu_2 \geq 0$. The dual problem is therefore

$$\max_{\lambda, \mu_1, \mu_2} \lambda$$

subject to

$$\mu_1 = 4 - \lambda,$$

$$\mu_2 = 2 + \lambda,$$

$$\mu_1 \geq 0,$$

$$\mu_2 \geq 0.$$

As the parameters μ_1 and μ_2 do not appear in the objective function, we can rewrite (1) and (2) as

$$4 - \lambda \geq 0,$$

$$2 + \lambda \geq 0,$$

and eliminate the μ 's, so that the dual problem becomes

$$\max_{\lambda} \lambda$$

subject to

$$4 - \lambda \geq 0,$$

$$2 + \lambda \geq 0.$$

The optimal solution is $\lambda^* = 4$. Note that $\mu_1^* = 4 - \lambda^* = 0$ and $\mu_2^* = 2 + \lambda^* = 6$. The optimal value is 4. It is the same value as the primal problem.